PCA :

$$\Sigma = P \Lambda P^T = \sum_{i=1}^{p} \lambda_i e_i e_i^T$$

$$= \begin{bmatrix} \sqrt{\lambda_1} e_1 & \cdots & \sqrt{\lambda_m} e_m \end{bmatrix} \begin{bmatrix} \text{—} \sqrt{\lambda_1} e_1^T \text{—} \\ \vdots \\ \text{—} \sqrt{\lambda_m} e_m^T \text{—} \end{bmatrix} + \begin{bmatrix} \sqrt{\lambda_{m+1}} e_{m+1} & \cdots & \sqrt{\lambda_p} e_p \end{bmatrix} \begin{bmatrix} \text{—} \sqrt{\lambda_{m+1}} e_{m+1}^T \text{—} \\ \vdots \\ \text{—} \sqrt{\lambda_p} e_p^T \text{—} \end{bmatrix}$$

$$= L_m L_m^T + L_{m+1} L_{m+1}^T = L L^T + \Psi \qquad (L = L_m)$$

$\Psi \to$ dig elems of $L_{m+1} L_{m+1}^T$

error matrix $= \Sigma - LL^T - \Psi$

sum of squares of entres of matrix $(\Sigma - LL^T - \Psi)$

Frobenous Norm

$\leq$ sum of sq of entres of the matrix $\Sigma - LL^T$

both dig and off dig errors are here

dig were taken care of

better fit model (error red)

$$= \text{sum of sq of entries of } \underbrace{L_{m+1} L_{m+1}^T}_{A}$$

$$= \text{trace}(AA^T)$$

$$= \text{trace}\left(L_{m+1} L_{m+1}^T \left(L_{m+1} L_{m+1}^T\right)^T\right)$$

$$= \text{trace}\left(P_2 \Lambda_2 P_2^T \; P_2 \Lambda_2 P_2^T\right)$$

$$= \text{trace}\left(P_2 \Lambda_2^2 P_2^T\right) \qquad [P_2^T P_2 = I]$$

$$\Sigma = P \Lambda P^T$$

$$P = \begin{bmatrix} e_1 \cdots e_m & \underbrace{e_{m+1} \cdots e_p}_{P_2} \end{bmatrix} \qquad \Lambda = \begin{bmatrix} \lambda_1 & & & & & \\ & \lambda_2 & & & & \\ & & \ddots & & & \\ & & & \lambda_m & & \\ & & & & \lambda_{m+1} & \\ & & & & & \ddots \\ & & & & & & \lambda_p \end{bmatrix}$$

Λ_2

P_2 dim $= p \times p - m$

$$\Lambda_2 = \begin{bmatrix} \lambda_{m+1} & & 0 \\ & \ddots & \\ 0 & & \lambda_p \end{bmatrix}$$

$P_2^T P_2 = I$

by ortho-normality of eigen vectors that makes colums of P

$$\text{trace}\left(P_2 \Lambda_2^2 P_2^T\right) = \text{trace}\left(\Lambda_2^2 \underbrace{P_2^T P_2}_{\to I}\right)$$

$$= \text{trace}\left(\Lambda_2^2\right)$$

$$= \sum_{i=m+1}^{p} \lambda_i^2$$

$$\Rightarrow \|\Sigma - LL^T - \Psi\|_F^2 \leq \sum_{i=m+1}^{p} \lambda_i^2$$

Frobenius norm

if ε_1 is error threshold

then $\sum_{j=m+1}^{p} \lambda_j^2 < \varepsilon_1$

optimal m_0?

$p = 5$ standardized data sample

R = co-relation matrix

$$= \begin{bmatrix} 1 & 0.02 & 0.96 & 0.42 & 0.01 \\ & 1 & 0.13 & 0.71 & 0.05 \\ & & 1 & 0.50 & 0.11 \\ & & & 1 & 0.79 \\ & & & & 1 \end{bmatrix}_{5\times 5}$$

$\hat{\lambda}_1 = 2.853$

$\hat{\lambda}_2 = 1.806$

$\hat{\lambda}_3 = 0.204$

$\hat{\lambda}_4 = 0.102$

$\hat{\lambda}_5 = 0.034$

total var of $R = 5$

$$\frac{\hat{\lambda}_1 + \hat{\lambda}_2}{5} * 100\% \simeq 93.18\%$$

let $m = 2$

$$\hat{\lambda}_3^2 + \hat{\lambda}_4^2 + \hat{\lambda}_5^2 \leq 0.06 = \varepsilon$$

$$\hat{e}_1 = \begin{bmatrix} 0.3316 \\ 0.4604 \\ 0.3822 \\ 0.5563 \\ 0.4729 \end{bmatrix} \qquad \hat{e}_2 = \begin{bmatrix} 0.6067 \\ -0.3897 \\ 0.556 \\ -0.078 \\ -0.4038 \end{bmatrix}$$

$$\hat{L} = \left[\sqrt{\hat{\lambda}_1}\,\hat{e}_1 \quad \sqrt{\hat{\lambda}_2}\,\hat{e}_2\right]_{2\times 5}$$

take $\hat{\Psi} = \text{dig}\left(R - \hat{L}\hat{L}^T \right)$

that will give factor modle

mathematical justification of:

$$\| \Sigma - LL^T - \Psi \|_F^2 \leq \| \Sigma - LL^T \|_F^2$$

$$\text{lhs} = \sum_{i=1}^{p} \sum_{j=1}^{p} \left(\sigma_{ij} - l_{ij} - \psi_{ij} \right)^2$$

$$= \sum_{i=1}^{p} \sum_{\substack{j=1 \\ (j \neq i)}}^{p} \left(\sigma_{ij} - l_{ij} \right)^2 + \sum_{\substack{i=1 \\ (j=i)}}^{p} \left(\sigma_{ii} - l_{ii} - \psi_i \right)^2$$